

The unit circle



- 1
- | | | | | | | | |
|---|------------|---|------------|---|------------|---|------------|
| a | Quadrant 4 | b | Quadrant 1 | c | Quadrant 2 | d | Quadrant 3 |
| e | Quadrant 1 | f | Quadrant 4 | g | Quadrant 3 | h | Quadrant 2 |
- 2
- | | | | | |
|---|---|-------------------|----|-------------------|
| a | i | Quadrants 1 and 2 | ii | Quadrants 3 and 4 |
| b | i | Quadrants 1 and 4 | ii | Quadrants 2 and 3 |
| c | i | Quadrants 1 and 3 | ii | Quadrants 2 and 4 |
- 3
- | | | | | | | | |
|---|----------|---|----------|---|----------|---|----------|
| a | Negative | b | Positive | c | Positive | d | Negative |
| e | Positive | f | Positive | g | Negative | h | Negative |
| i | Negative | | | | | | |
- 4
- | | | | | | | | |
|---|-------|---|-------|---|------|---|------|
| a | False | b | False | c | True | d | True |
| e | True | | | | | | |
- 5
- | | | | |
|---|------------|---|------------|
| a | Quadrant 1 | b | Quadrant 4 |
|---|------------|---|------------|
- 6
- | | | | | | | | |
|---|------------|---|------------|---|------------|---|------------|
| a | Quadrant 2 | b | Quadrant 2 | c | Quadrant 2 | d | Quadrant 1 |
|---|------------|---|------------|---|------------|---|------------|

Equivalent angles

7 30°

- 8
- | | | | | | | |
|---|---|-----------------|----|------------------|-----|------------------|
| a | i | $Q(-a, b)$ | ii | $R(-a, -b)$ | iii | $S(a, -b)$ |
| b | i | $\sin 49^\circ$ | ii | $-\sin 49^\circ$ | iii | $-\sin 49^\circ$ |
| c | | | | | | |



9

a i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b i $-\cos 63^\circ$ ii $-\cos 63^\circ$ iii $\cos 63^\circ$

c i $-\cos x$ ii $-\cos x$ iii $\cos x$

10

a i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b i $-\tan 62^\circ$ ii $\tan 62^\circ$ iii $-\tan 62^\circ$

c i $-\tan x$ ii $\tan x$ iii $-\tan x$

11

a i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$ b i $\frac{20}{29}$ ii $-\frac{21}{29}$ iii $-\frac{20}{21}$

c i $-\frac{21}{29}$ ii $\frac{20}{29}$ iii $-\frac{21}{20}$

12

a 65° b 30° c 60° d 75°

13

a $\sin 87^\circ$ b $-\cos 15^\circ$ c $-\tan 61^\circ$ d $-\sin 41^\circ$

e $-\cos 42^\circ$ f $\tan 71^\circ$

14

a $\sin \theta$ b $\cos \theta$ c $\sin \theta$ d $-\cos \theta$

e $-\sin \theta$ f $\cos \theta$ g $-\sin \theta$ h $\cos \theta$

15

a 0.93 b 0.36 c -0.93 d 0.36

e -0.36 f -0.93 g 0.36 h -0.93

16

a 0.56

b 0.49



c 0.89

17

a $P(0.41, 0.91)$

b $(-0.41, 0.91)$

c 0.91

d -0.41

e -2.25

18

a $\sin \theta = -\frac{4}{5}$

b $\tan \theta = -\frac{4}{3}$

19

a $\cos \theta = \pm \frac{3}{4}$

b $\tan \theta = \pm \frac{\sqrt{7}}{3}$

20

a $\sin \theta = \frac{3}{5}$

b $\cos \theta = -\frac{4}{5}$

c $\tan \theta = -\frac{3}{4}$

21

a $\cos \theta = \frac{10}{\sqrt{269}}$

b $\sin \theta = -\frac{13}{\sqrt{269}}$

22 $\cos \theta = -\frac{8}{17}$

23

a $\sin \theta = \frac{11}{61}$

b $\sin \theta = \frac{\sqrt{13}}{7}$

24

a $\tan \theta = \frac{2\sqrt{10}}{3}$

b $\tan \theta = \frac{1}{3}$



Exact trigonometric ratios

25

a

i $\frac{\sqrt{3}}{2}$

ii $\frac{1}{2}$

iii $\sqrt{3}$

b

i 120°

ii 240°

iii 300°

c

i $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

ii $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

iii $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

26

a

i $\frac{1}{2}$

ii $\frac{\sqrt{3}}{2}$

iii $\frac{1}{\sqrt{3}}$

b

i 150°

ii 210°

iii 330°

c

i $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

ii $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

iii $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

27

a 30°

b $-\frac{\sqrt{3}}{2}$

28

a $-\frac{1}{\sqrt{2}}$

b $-\frac{1}{\sqrt{2}}$

c 1

d $-\frac{1}{2}$

e $\frac{1}{\sqrt{2}}$

f $\frac{1}{\sqrt{2}}$

g $-\frac{1}{\sqrt{3}}$

h $-\frac{\sqrt{3}}{2}$

i $\frac{\sqrt{3}}{2}$

j $-\frac{1}{\sqrt{3}}$

k 0

l $\sqrt{3} - 1$

m $\frac{15}{4}$

29 $-\frac{1}{4}$